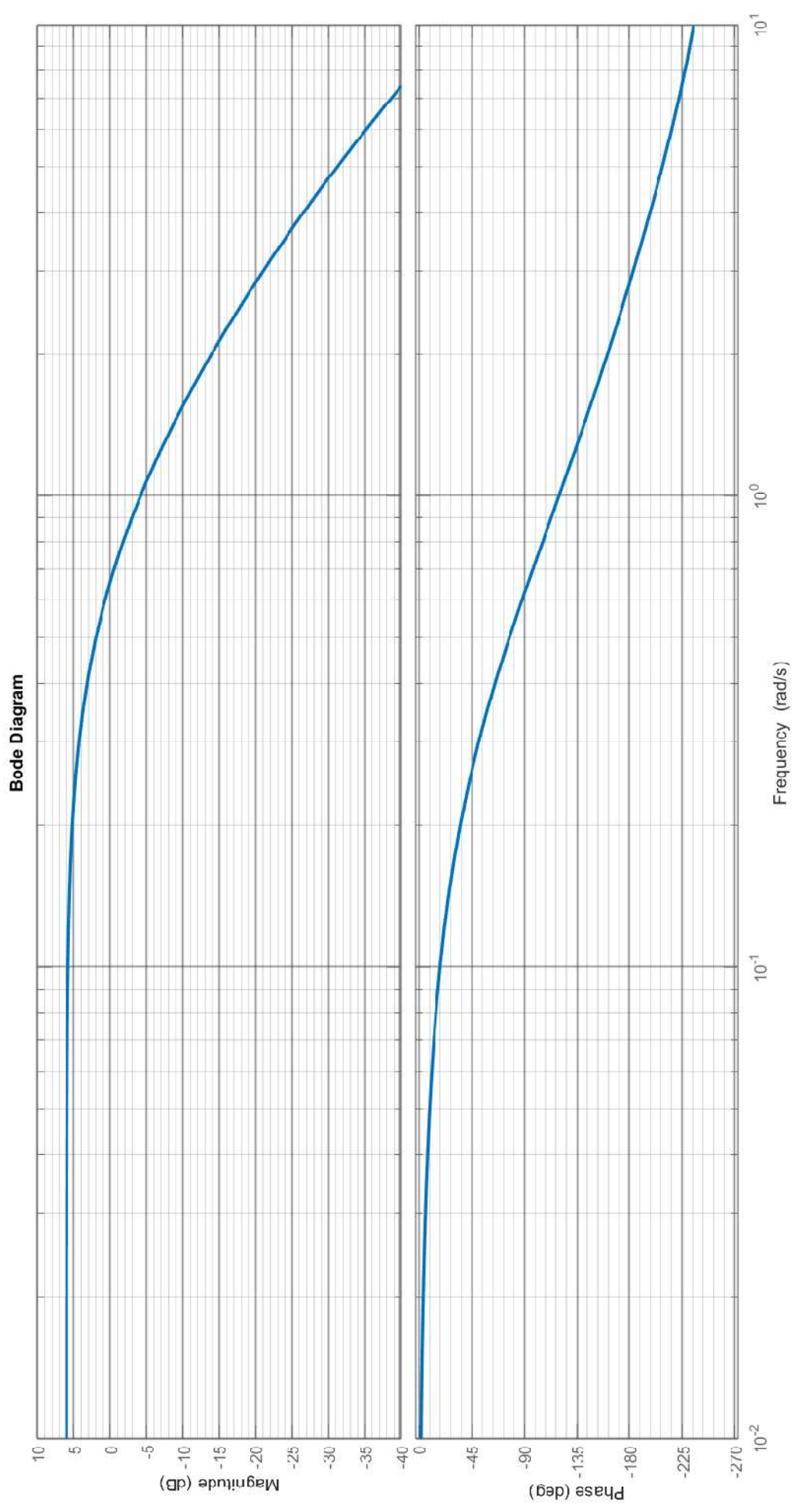


# Tut Test 5 [15 marks]

The Bode plot of a system is shown on the next page. The system is stable in open loop, and will be placed in negative feedback with a proportional controller with the aim of improving its performance to meet the following specifications:

- i. Tracking of constant setpoints with less than 2% accuracy.
  - ii. Overshoot of less than 10%.
  - iii. Sensitivity of less than 6 dB at all frequencies.
  - iv. Sensitivity of less than 2 dB at 3 rad/s.
- 
- a) Sketch the frequency response of the system on the inverse Nichols chart provided and indicate boundaries for each of these specifications. [5 marks]
  - b) For each specification, determine the range of gains for which it will be satisfied. You may state your answers in decibels. [4 marks]
  - c) What is the best possible accuracy with respect to position setpoints without violating specifications ii-iii? [2 marks]
  - d) Is the accuracy specification achievable for this system even if the other specifications are ignored? If not, give the best achievable accuracy. [3 marks]

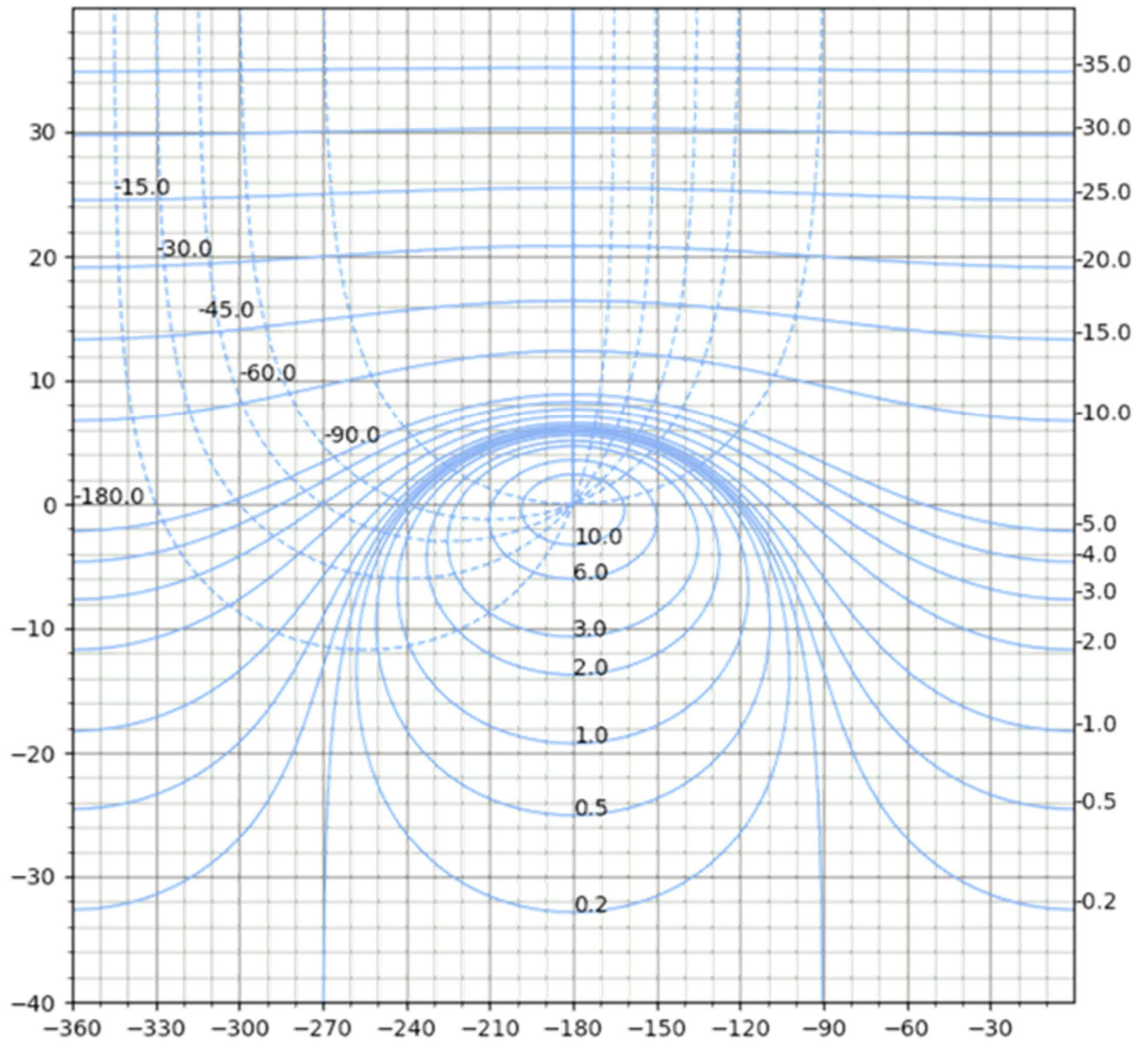


# Tut Test 5

Name: \_\_\_\_\_

Answer sheet

Student No: \_\_\_\_\_



a)

b)

Student No: \_\_\_\_\_

c)

d)